SURYAGRID AI

Business Proposal and RFP Brief

Multi-horizon renewable forecast system for deviation, dispatch, reserve and day-ahead planning

Document Type	Business Proposal / RFP Brief
Version	v1.0
Use	Sample PDF upload for LLM requirement extraction, planning and forecast design
Data Status	Synthetic demonstration - not real plant SCADA or commercial data
Primary Objective	Enable a model to infer business use case, horizon, variables, methods, metrics and data gaps

LLM extraction note: This document intentionally uses clear labels, tables and repeated field names so a document-reading model can extract business objectives, SCADA/MET/NWP variables, forecast horizons, metrics and RFP requirements with minimal ambiguity.

1. Executive Summary

This proposal describes a multi-horizon, uncertainty-aware renewable forecasting system for a large solar plant portfolio. The system is designed to convert business objectives such as deviation penalty reduction, grid dispatch support, reserve planning and ramp warnings into a concrete forecast plan: horizons, variables, model families, metrics and operational actions.

The proposed platform uses SCADA, meteorological observations, NWP ensemble forecasts, grid constraints, schedule data and historical event labels. It does not assume one universal algorithm. Each business use case is mapped to the forecast horizon and method family that best supports the operational decision.

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LLM-readable requirement block: BUSINESS_OBJECTIVE = minimize_deviation_penalties + support_grid_dispatch + real_time_ramp_warning + day_ahead_uncertainty. TARGET_ASSET = Pavagada Solar Park synthetic sample. REQUIRED_OUTPUT = horizon_plan + data_requirements + model_selection + metrics + RFP_requirements.
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2. Business Drivers and Operational Use Cases

Business use case	Operational decision	Recommended horizons	Why forecast is needed
Real-time ramp warning	Prepare control-room action before output changes quickly.	5min, 15min	Fast ramps can cause schedule deviation, reserve stress and operator surprise.
Deviation penalty reduction	Compare forecast, schedule and actual; update dispatch blocks.	15min, 30min, 1hour, 2hour	Deviation risk is a business objective, so forecast must link to schedule and penalties.
Intraday reserve planning	Estimate reserve requirement and ramp risk later today.	2hour, 4hour	Reserve decisions need uncertainty-aware bands, not only a single point forecast.
Day-ahead scheduling	Submit or revise tomorrow schedule with uncertainty bands.	day_ahead_40hour	Weather forecast uncertainty dominates day-ahead output risk.
Weekly planning outlook	Plan maintenance and low-generation risk windows.	8day	Exact 5-minute output is unrealistic; broad regime risk is more useful.
Curtailment diagnosis	Separate grid-limit clipping from weather/model error.	5min to day_ahead_40hour	Curtailment must not be penalized as forecast failure.

3. Forecast Solution Design

Horizon	Business name	Method family	Uncertainty approach	Primary metrics
5min	Real-time nowcast	Persistence + ramp-rate correction + SCADA quality gate	Empirical residual band	rolling nMAE, ramp hit rate, lead time, phase error, SCADA freshness
15min/30min	Intra-hour dispatch correction	Persistence + physical weather-to-power blend	Empirical residual + event widening	nMAE, coverage, false alarm rate, amplitude error
2hour/4hour	Intraday reserve planning	Physical weather-to-power + Gaussian residual	Gaussian residual band / quantile band	RMSE, nMAE, bias, P10-P90 coverage, ramp amplitude error
day_ahead_40 hour	Day-ahead probabilistic forecast	NWP ensemble weather-to-power + quantile calibration	Calibrated ensemble quantiles	MAE, RMSE, bias, reliability, rank histogram, coverage
8day	Medium-range planning outlook	Scenario ensemble + analog days + climatology band	Scenario band	regime accuracy, low-generation risk hit, weekly coverage, scenario spread

4. Required Data Categories

Data category	Minimum fields	Purpose
SCADA	actual_mw, possible_power_mw, scheduled_mw, local_limit_mw, control_setpoint_mw, availability_pct, scada_quality, scada_latency_seconds, ramp_rate_mw_min	Live nowcast, schedule deviation, curtailment separation and quality gating.
Solar met observations	poa_wm2, ghi_wm2, ambient_temp_c, module_temp_c, cloud_cover_pct, cloud_obstruction_probability, sky_camera_cloud_motion	Weather-to-power conversion and early cloud-ramp warning.
NWP ensemble	nwp_ghi_forecast_wm2, nwp_cloud_cover_pct, nwp_temperature_c, nwp_ensemble_member_id, nwp_issue_time, nwp_valid_time	Day-ahead and medium-range probabilistic forecasts.

Data category	Minimum fields	Purpose
Grid/market	demand_forecast_mw, reserve_requirement_mw, deviation_penalty_rate, grid_constraint_flag, dispatch_block_id	RFP and business objective evaluation; reserve and demand context.
Event labels	event_type, root_cause, start_time, end_time, forecast_start_time, actual_start_time, max_loss_mw	Historical validation, ramp hit rate, curtailment precision and operator explanation.

5. Agentic Workflow

- Ingest business request, plant brief, SCADA specification and available data catalog.
- Infer business objective: e.g., deviation penalty, reserve, day-ahead schedule, ramp warning, curtailment diagnosis.
- Select horizon plan: 5min, 15min, 2hour, day-ahead 40hour, 8day as needed.
- Map each horizon to required variables, model family, uncertainty method and evaluation metrics.
- Report data gaps and clarify whether missing fields block go-live or only reduce confidence.
- Generate RFP-style requirements and validation acceptance tests.
- During simulation, update model health and event panels by grid, plant, hub and block level.

6. RFP-Style Functional Requirements

Requirement ID	Requirement statement	Acceptance evidence
FR-01	System shall produce 5min nowcast using recent SCADA, ramp-rate and SCADA quality gate.	Live replay shows actual MW, forecast P50, uncertainty band and event markers.
FR-02	System shall produce day-ahead P10/P50/P90 forecasts using NWP ensemble inputs and calibration.	Reliability diagram, rank histogram and coverage statistics are displayed.
FR-03	System shall separate curtailment from weather forecast error using possible power and local limit.	Curtailment events show actual clipped near local_limit while possible_power is higher.
FR-04	System shall show hierarchy-level outputs at grid, plant, hub and block level.	User can select level and see relevant events, variables and model technique.
FR-05	System shall generate plain-language explanations for horizon, variables, metrics and actions.	Output JSON includes plain_language_explanation, missing_information and data_gap_impact.

7. Business Proposal Success Metrics

Metric	Plain-language meaning	Target / interpretation
rolling_nMAE	Recent average forecast miss as percent of capacity.	< 5% for 5min nowcast target in demo setting.
P10-P90 coverage	How often actual output falls inside uncertainty band.	Should be near expected interval; low coverage means band is too narrow.
Ramp hit rate	How many actual ramp events were detected in time.	Higher is better; evaluate with lead time.
Bias	Systematic over-forecast or under-forecast.	Near zero; persistent bias triggers recalibration.
Reliability	Probability honesty for probabilistic forecasts.	Forecast probabilities should match observed frequencies.

8. LLM Extraction Checklist

When this PDF is uploaded, the model should extract: business use cases, horizon plan, data variables, model families, uncertainty methods, metrics, RFP requirements, missing information and clarification questions. If the model cannot find a variable, it must not invent it; it should mark the field as missing.